

Errors in transfer learning using resnet101

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KEN SUEMATSU on 15 Mar 2021

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Commented: Tan on 14 May 2023

Accepted Answer: Akira Agata **STAFF** **EDITOR**

I would like to use resnet101 to do transfer learning.

When I build the network and use the trainNetwork function as shown below, I get the following error. What is the cause?

Layer 'res2a': unconnected input. The input of each layer must be coupled with the output of another layer.

An unconnected input was detected:

```
net = resnet101;
layers = net.Layers;

layers = [
    layers(1:344)
    fullyConnectedLayer(NumberOfclasses)
    layers(346)
    classificationLayer];

options = trainingOptions('sgdm',...
    'MiniBatchSize',16,...
    'InitialLearnRate', 0.0001, ...
    ...);

trainNetwork(TrainImage,TrainData,layers,options);
```

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✓ Accepted Answer



Akira Agata **STAFF** **EDITOR**
on 18 Mar 2021

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Since ResNet-101 is imported as a DAGNetwork object, the following steps will be needed (more details can be found in this [Link](#))

1. Convert DAGNetwork object to LayerGraph object
2. Replace the last few layers
3. Freeze bias/weight of initial layers (optional)
4. Re-connect all the layers in the original order by using the support function `createLgraphUsingConnections`

So the MATLAB code will be like this.

```
net = resnet101;
```

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```
% 1. Convert DAGNetwork object to LayerGraph object
lgraph = layerGraph(net);

% 2. Replace the last few layers
lgraph = replaceLayer(lgraph, 'fc1000', ...
    fullyConnectedLayer(NumberOfClasses, 'Name', 'fcNew'));
lgraph = replaceLayer(lgraph, 'ClassificationLayer_predictions', ...
    classificationLayer('Name', 'ClassificationNew'));

% 4. Re-connect all the layers in the original order
% by using the support function createLgraphUsingConnections
layers = lgraph.Layers;
connections = lgraph.Connections;
lgraph = createLgraphUsingConnections(layers, connections);

% Train the network
options = trainingOptions('sgdm', ...
    'MiniBatchSize', 16, ...
    'InitialLearnRate', 0.0001, ...
    ...);
net = trainNetwork(imdsTrain, lgraph, options);
```

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baby on 18 Feb 2022



I dont think its an error, its a warning but somehow it appears in red. You can ignore this and proceed with the training procedure, and to make sure you can use command:

```
analyzeNetwork(lgraph)
```

if found no error, the training will process very nicely. (I hope).



Tan on 14 May 2023



hi although using the command

```
layers = lgraph.Layers;
connections = lgraph.Connections;
lgraph = createLgraphUsingConnections(layers, connections);
```

it also show the same error:

```
trainedNet = trainNetwork(augmentedTrainingSet, lgraph, options);
```

Error using trainNetwork

Invalid network.

Caused by:

Layer 'inception_3a-output': Unconnected input. Each layer input must be connected to the output of another layer.

Layer 'inception_3b-output': Unconnected input. Each layer input must be connected to the output of another layer.

Layer 'inception_4a-output': Unconnected input. Each layer input must be connected to the output of another layer.

Layer 'inception_4b-output': Unconnected input. Each layer input must be connected to the output of another layer.

Layer 'inception_4c-output': Unconnected input. Each layer input must be connected to the output of another layer.

Layer 'inception_4d-output': Unconnected input. Each layer input must be connected to the output of another layer.

Layer 'inception_4e-output': Unconnected input. Each layer input must be connected to the output of another layer.

Layer 'inception_5a-output': Unconnected input. Each layer input must be connected to the output of another layer.

Layer 'inception_5b-output': Unconnected input. Each layer input must be connected to the output of another layer.

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Error using trainNetwork. Invalid network.

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Invalid training data. The output size (1000) of the last layer does not match the number of classes (5).

2 Answers

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